

Operating Systems Lab All Tasks

Name :Sayan Ali

Roll No: 2401010163

Section :B

Task 1 – Display the current working directory

Command:

`pwd`

Explanation: Displays the complete path of the current working directory.

Output/Screenshot:

```
sayan@ubuntu:~/sayan$ pwd
/home/sayan/sayan
```

Task 2 – Display the contents of the current directory

Command:

`ls`

Explanation: Lists files and directories in the current directory.

Output/Screenshot:

```
sayan@ubuntu:~/sayan$ ls
2.txt  this
```

Task 3 – Display all files, including hidden files

Command:

`ls -a`

Explanation: Displays all files and directories, including hidden files.

Output/Screenshot:

```
sayan@ubuntu:~/sayan$ ls -a
.  ..  2.txt  this
```

Task 4 – Create Linux_Assignment

Command:

```
mkdir Linux_Assignment
```

Explanation: Creates the Linux_Assignment directory.

Output/Screenshot:

```
sayan@ubuntu:~/sayan$ mkdir Linux_Assignment
```

Task 5 – Enter Linux_Assignment

Command:

```
cd Linux_Assignment
```

Explanation: Changes to the Linux_Assignment directory.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop$ cd Linux_Assignment
```

Task 6 – Create four directories

Command:

```
mkdir Documents Programs Reports Backup
```

Explanation: Creates the four required directories.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment$ mkdir Documents Programs Reports Backup
sayan@ubuntu:~/Desktop/Linux_Assignment$ ls
```

Task 7 – Verify the directories

Command:

```
ls
```

Explanation: Displays and verifies Documents, Programs, Reports, and Backup.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment$ ls
Backup  Documents  Programs  Reports
```

Task 8 – Create three files

Command:

cd Documents

touch file1.txt file2.txt file3.txt

Explanation: Enters Documents and creates the three text files.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment$ cd Documents
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ touch file1.txt file2.txt file3.txt
```

Task 9 – Display the files

Command:

ls

Explanation: Lists the files in Documents.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ ls
file1.txt  file2.txt  file3.txt
```

Task 10 – Write text into file1.txt

Command:

echo "Operating Systems" > file1.txt

echo "Linux Command Practice" >> file1.txt

echo "KRMU" >> file1.txt

Explanation: Writes the required text into file1.txt.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ echo "Linux Command Practice" >> file1.txt
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ echo "KRMU" >> file1.txt
```

Task 11 – Display file1.txt

Command:

cat file1.txt

Explanation: Displays the contents of file1.txt.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ cat file1.txt  
Linux Command Practice
```

Task 12 – Copy file1.txt to Backup

Command:

```
cp file1.txt ../Backup/
```

Explanation: Copies file1.txt to Backup.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ cp file1.txt ../Backup/
```

Task 13 – Rename file2.txt

Command:

```
mv file2.txt student.txt
```

Explanation: Renames file2.txt to student.txt.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ mv file2.txt student.txt
```

Task 14 – Move file3.txt to Reports

Command:

```
mv file3.txt ../Reports/
```

Explanation: Moves file3.txt to Reports.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ mv file3.txt ../Report/  
mv: cannot move 'file3.txt' to '../Report/': Not a directory
```

Task 15 – Verify Backup

Command:

```
ls ../Backup
```

Explanation: Displays Backup and verifies file1.txt.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ ls ../Backup  
file1.txt
```

Task 16 – Count lines, words and characters

Command:

```
wc file1.txt
```

Explanation: Displays the counts for file1.txt.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ wc file1.txt
1  3 23 file1.txt
```

Task 17 – First few lines

Command:

```
head file1.txt
```

Explanation: Displays the beginning of file1.txt.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ head file1.txt
Linux Command Practice
```

Task 18 – Last few lines

Command:

```
tail file1.txt
```

Explanation: Displays the end of file1.txt.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ tail file1.txt
Linux Command Practice
```

Task 19 – Search for Linux

Command:

```
grep "Linux" file1.txt
```

Explanation: Searches for the word Linux in file1.txt.

Output/Screenshot:

```
sayan@ubuntu:~$ grep "Linux" file1.txt
```

Task 20 – Find all .txt files

Command:

```
cd ..  
find . -name "*.txt"
```

Explanation: Finds all .txt files under Linux_Assignment.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ cd  
sayan@ubuntu:~/Desktop/Linux_Assignment$ find -name "*.txt"  
./Reports/file3.txt  
./Backup/file1.txt  
./Documents/student.txt  
./Documents/file1.txt
```

Task 21 – Display permissions

Command:

```
cd Documents  
ls -l
```

Explanation: Displays detailed file permissions.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment$ cd Documents  
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ ls -l  
total 4  
-rw-rw-r-- 1 sayan sayan 23 Sep 17 06:55 file1.txt  
-rw-rw-r-- 1 sayan sayan  0 Sep 17 06:55 student.txt
```

Task 22 – Change permissions

Command:

```
chmod 644 file1.txt
```

Explanation: Owner gets read/write; group and others get read-only.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ chmod 644 file1.txt
```

Task 23 – Verify permissions

Command:

```
ls -l file1.txt
```

Explanation: Verifies the permission setting.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ ls -l file1.txt  
-rw-r--r-- 1 sayan sayan 23 Sep 17 06:55 file1.txt
```

Task 24 – Explain -rwxr-xr--

Command:

```
- rwX r-x      r--
```

Owner Group Others

Numeric form: 754

Explanation: Regular file; owner has rwX, group has r-x, others have r--.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ ls -l
total 4
-rw-rw-r-- 1 sayan sayan 23 Sep 17 06:55 file1.txt
-rw-rw-r-- 1 sayan sayan  0 Sep 17 06:55 student.txt
```

Task 25 – uname -a

Command:

```
uname -a
```

Explanation: Displays detailed Linux system and kernel information.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ uname -a
Linux ubuntu 7.0.0-31-generic #31-Ubuntu SMP PREEMPT_DYNAMIC Sat Aug 1 04:26:38 UTC 2026 x86_64 GNU/Linux
```

Task 26 – whoami

Command:

```
whoami
```

Explanation: Displays the current username.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ whoami
sayan
```

Task 27 – date

Command:

```
date
```

Explanation: Displays the current date and time.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ date
Thu Sep 17 07:08:37 AM UTC 2026
```

Task 28 – df -h

Command:

df -h

Explanation: Displays disk usage in human-readable form.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            1.6G  3.3M  1.6G   1% /run
/dev/sda2        18G   6.8G  9.6G  42% /
tmpfs            4.0G   0    4.0G   0% /dev/shm
tmpfs            4.0G  32K   4.0G   1% /tmp
none             1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
none             1.0M   0    1.0M   0% /run/credentials/systemd-resolved.service
tmpfs            803M  128K  803M   1% /run/user/1000
```

Task 29 – free -h

Command:

free -h

Explanation: Displays RAM and swap usage.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.8Gi        1.5Gi        4.6Gi          50Mi        2.1Gi        6.3Gi
Swap:          0B           0B           0B
```

Task 30 – uptime

Command:

uptime

Explanation: Displays system uptime and load information.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ uptime
07:08:57 up 58 min,  1 user,  load average: 1.05, 0.53, 0.39
```

Task 31 – Display running processes

Command:

ps

Explanation: Displays processes associated with the current terminal.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ ps
  PID TTY          TIME CMD
 14714 pts/1        00:00:00 bash
 15391 pts/1        00:00:00 ps
```

Task 32 – Detailed processes

Command:

```
ps aux
```

Explanation: Displays detailed information about system processes.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ ps aux
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.5	0.2	26572	17428	?	Ss	06:10	0:20	/usr/lib/systemd/systemd
root	2	0.0	0.0	0	0	?	S	06:10	0:00	[kthreadd]
root	3	0.0	0.0	0	0	?	I<	06:10	0:00	[pool_workqueue_release]
root	4	0.0	0.0	0	0	?	I<	06:10	0:00	[kworker/R-rcu_gp]
root	17	0.1	0.0	0	0	?	S	06:10	0:04	[rcu_exp_gp_kthread_worke
root	18	0.0	0.0	0	0	?	S	06:10	0:00	[migration/0]
sayan	14714	0.0	0.1	10840	5230	pts/1	Ss	07:00	0:00	bash

Task 33 – Start a process

Command:

```
sleep 100
```

Explanation: Starts a process that sleeps for 100 seconds.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ sleep 100 &
[1] 15410
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ pgrep sleep
15410
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ kill 15410
[1]+  Terminated                  sleep 100
```

Task 34 – Find PID

Command:

`pgrep sleep`

Explanation: Displays the PID of the sleep process.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ pgrep sleep
15410
```

Task 35 – Terminate sleep process

Command:

`kill <PID>`

Explanation: Terminates the process using its PID; replace <PID> with the actual PID.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ kill 15410
```

Task 36 – Verify termination

Command:

`pgrep sleep`

Explanation: No output means the sleep process has terminated.

Output/Screenshot:

```
sayan@ubuntu:~$ pgrep sleep
```

Task 37 – Create students.txt

Command:

`printf "Aman\nRiya\nRahul\nPriya\nNeha\nKaran\n" > students.txt`

Explanation: Creates students.txt with the six required names.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ printf "Aman\nRiya\nRahul\nPriya\nNeha\nKaran\n" > students.txt
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ ls students.txt
students.txt
```

Task 38 – Display students.txt

Command:

`cat students.txt`

Explanation: Displays all student names.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ cat students.txt
Aman
Riya
Rahul
Priya
Neha
Karan
```

Task 39 – Count students

Command:

```
wc -l students.txt
```

Explanation: Counts the six student-name lines.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ wc -l students.txt
6 students.txt
```

Task 40 – Sort

alphabetically

Command:

```
sort students.txt
```

Explanation: Sorts the names alphabetically.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ sort students.txt
Aman
Karan
Neha
Priya
Rahul
Riya
```

Task 41 – Reverse order

Command:

```
sort -r students.txt
```

Explanation: Displays names in reverse alphabetical order.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ sort -r students.txt
Riya
Rahul
Priya
Neha
Karan
Aman
```

Task 42 – Use a pipe

Command:

```
cat students.txt | wc -l
```

Explanation: Sends cat output through a pipe to wc -l.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ cat students.txt | wc -l
6
```

Task 43 – Create OS_Lab structure

Command:

```
cd ~
```

```
mkdir -p OS_Lab/Programs/C
```

```
mkdir -p OS_Lab/Programs/Python
```

```
mkdir -p OS_Lab/Assignments
```

```
mkdir -p OS_Lab/Backup
```

Explanation: Creates the required OS_Lab structure.

Output/Screenshot:

```
sayan@ubuntu:~/Desktop/Linux_Assignment/Documents$ cd
sayan@ubuntu:~$ mkdir -p OS_Lab/Programs/C
sayan@ubuntu:~$ mkdir -p OS_Lab/Assignments
sayan@ubuntu:~$ mkdir -p OS_Lab/Backup
sayan@ubuntu:~$ find OS_Lab -type d
OS_Lab
OS_Lab/Assignments
OS_Lab/Programs
OS_Lab/Programs/C
OS_Lab/Backup
```

Task 44 – Create Python files

Command:

```
cd ~/OS_Lab/Programs/Python
```

```
touch hello.py addition.py calculator.py
```

Explanation: Creates the three Python files.

Output/Screenshot:

```
sayan@ubuntu:~$ cd ~/OS_Lab/Programs/Python
bash: cd: /home/sayan/OS_Lab/Programs/Python: No such file or directory
sayan@ubuntu:~$ touch hello.py addition.py calculator.py
sayan@ubuntu:~$ ls
Desktop    Downloads  Music      Public     Videos    addition.py  hello.py
Documents  OS_Lab    Pictures   Templates  snap       calculator.py sayan
```

Task 45 – Write and run hello.py

Command:

```
printf 'print("Hello Linux")\nprint("Welcome to Operating Systems Lab")\n' > hello.py
hello.py
python3 hello.py
```

Explanation: Creates and runs the required Python program.

Output/Screenshot:

```
sayan@ubuntu:~$ printf 'print("Hello Linux")\nprint("Welcome to Operating Systems Lab")\n' > hello.py
sayan@ubuntu:~$ cat hello.py
print("Hello Linux")
print("Welcome to Operating Systems Lab")
```

Task 46 – Backup OS_Lab

Command:

```
cd ~
tar -czvf OS_Lab_Backup.tar.gz OS_Lab
```

Explanation: Creates a compressed backup archive of OS_Lab.

Output/Screenshot:

```
sayan@ubuntu:~$ cd
sayan@ubuntu:~$ tar -czvf OS_Lab_Backup.tar.gz OS_Lab
OS_Lab/
OS_Lab/Assignments/
OS_Lab/Programs/
OS_Lab/Programs/C/
OS_Lab/Backup/
```

Task 47 – Display OS_Lab size

Command:

```
du -sh OS_Lab
```

Explanation: Displays the total size of OS_Lab in human-readable format.

Output/Screenshot:

```
sayan@ubuntu:~$ du -sh OS_Lab
20K    OS_Lab
```

VIVA QUESTIONS – ANSWERS

1. What is a PID?

PID stands for Process ID. It is the numerical identifier assigned to a running process.

2. What is the purpose of ps?

ps displays information about running processes.

3. What is the purpose of kill?

kill sends a signal to a process, commonly requesting it to terminate.

4. What is a pipe (|)?

A pipe sends the output of one command as the input to another command.

5. What is input/output redirection?

It changes where a command gets input from or sends output to, such as a file.

6. Difference between > and >>?

> creates or overwrites a file, while >> appends content to the end of a file.

7. Why is python3 commonly used instead of python in Ubuntu?

python3 explicitly runs Python 3 and avoids ambiguity about which Python version is being used.